



Course Description	
COURSE NUMBER and NAME	MTH 306 Applied Calculus
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	This course introduces the foundational principles of applied calculus, encompassing key topics such as limits, derivatives, and integrals. As students progress, they will explore more advanced concepts and their practical applications in real-world scenarios. The course includes opportunities to solve practical problems using Python programming, enhancing students' comprehension of calculus theory while providing valuable hands-on experience.
REQUIRED TEXT	Stewart J. (2020). <i>Calculus: Early transcendentals</i> (9th ed.). Cengage Learning. ISBN-10: 1337613924 ISBN-13: 978-1337613927
INSTRUCTIONAL METHOD	Online / On-Campus



Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism (Rubric)	150	15%	1, 2, 3, 4, 5
Week 1 Discussion	50	5%	1, 2, 3
Week 1 Assignment	100	10%	1, 3
Week 2 Assignment	100	10%	1, 3
Week 3 Discussion	50	5%	1,2,3,5
Week 4 Assignment	100	10%	1, 3
Week 5 Assignment	100	10%	1,2,3,5
Week 6 Discussion	50	5%	1, 2, 3, 5
Week 6 Assignment	100	10%	1, 3
Week 7 Assignment	100	10%	1, 3, 4, 5
Week 8 Exam	100	10%	5
Total Points/Percentage	1000 Points	100%	



Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Wednesday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Saturday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines, and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the Virtual Class Session (VCS) submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Apply the principles of applied calculus to solve problems involving limits, derivatives, integrals, and their subtopics.	4, 6
2. Identify various theories and optimization techniques in calculus to ensure their appropriate application.	1, 6
3. Solve calculus problems through hands-on exercises.	3, 7
4. Utilize a programming language to solve calculus-related problems.	1, 3, 4
5. Relate applied calculus concepts to solve real world problems.	5, 6



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading: Calculus: Early Transcendentals (p. 77-172)
 - Introduction to Calculus
 - The limit of a function
 - Calculating limits using limit laws
 - Understanding One-Sided Limits graphically.
 - Continuity of Functions
 - First principle of Derivatives
 - Application of Limits: Tangent Lines and Rates of Change
- Videos:
 - [What is Calculus?](#)
 - [Introduction to Limits](#)
 - [Determine Limits Using Limit Laws \(Properties\)](#)
 - [Continuity using Limits](#)
 - [Application of the Limit Definition of the Derivative \(Quadratic Height Function\)](#)

Week 1 Discussion ([Rubric](#))

Discuss your understanding of calculus and how they can be applied to solve real-world problems. Elaborate on specific examples illustrating how calculus principles are utilized to address practical challenges.

Week 1 Assignment ([Rubric](#))

In this assignment, students need to solve the following problems using pen and paper.

- Use the first principle of derivatives to find the derivative of $g(x) = 3x^2 + 4x - 2$. Show each step of the calculation.
- The position function of an object moving along a straight line is given by $s(t) = 2t^2 - 3t + 1$. Find the velocity of the object at time $t = 2$ using the concept of limits and tangent lines.
- The population of a city is modeled by $P(t) = 1200e^{(0.02t)}$, where t is the time in years. Find the rate at which the population is growing after 3 years. Interpret the result in the context of the problem.



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Submission guidelines: Scan your completed solutions, compile them, and upload them to the portal in PDF format.



Week 2

Assignments to complete this week:

- Reading: Calculus: Early Transcendentals (p. 173-278)
 - Definition of Derivative
 - Basic Differentiation Rules
 - Product, Quotient, and Chain Rules
 - Differentiation of trigonometric function
 - Differentiation of exponential and logarithmic functions
 - Double differentiation
 - Implicit Differentiation
- Videos:
 - [Ex 1: Basic Derivatives Using the Power Rule](#)
 - [Derivatives with Function Notation: Product, Quotient, and Chain Rule Concept Check](#)
 - [Ex 1: Derivative of Trigonometric Functions -- Simplify Before Differentiating](#)
 - [Ex 1: Derivatives of Exponential Functions](#)
 - [The Derivative of A Natural Log Function Using Properties of Logarithms](#)
 - [Implicit Differentiation](#)

Week 2 Live Class Activities ([Rubric](#))

During a live class activity, students will be grouped according to class size. The following steps will then be carried out:

- The class instructor will allocate a specific amount of time to each group to prepare questions within the scope of the course lectures, focusing on topics such as basic differentiation, product/quotient/chain rules, and derivatives of trigonometric, exponential, logarithmic, and implicit differentiation. The instructor will provide guidance on the number of questions to be prepared.
- Subsequently, each group will be tasked with presenting their questions to another randomly selected group. The instructor will facilitate this process.
- The receiving group will then solve the problems provided by the presenting group.

Once the solutions are completed, each group will share their solutions with the entire live class.

Week 2 Assignment ([Rubric](#))

For this assignment, students are required to solve questions by using pen and paper numbered 1 to 10 and 61 to 70 from the review exercise of chapter 3 found on pages 270-271 of the textbook.



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Week 3

Assignments to complete this week:

- Reading: Calculus: Early Transcendentals (p. 279-370)
 - Derivative of parametric functions
 - Extrema on an Interval
 - Rolle's Theorem and the Mean Value Theorem
 - Increasing and Decreasing Functions and Concavity
 - Optimization Problems
 - Applications of Derivatives in Economics and Science
- Videos:
 - [The Derivative of Parametric Equations](#)
 - [Ex 1: Determine Relative Extrema Using The First Derivative Test](#)
 - [The Mean Value Theorem](#)
 - [Ex: Determine Increasing/Decreasing/Concavity Intervals of a Function](#)
 - [Ex 2: Max / Min Application Problem - Max Volume of a Box](#)
 - [Ex: Derivative Application - Maximize Profit](#)

Week 3 Live Class Activities ([Rubric](#))

During this live class activity, students will engage in a hands-on exercise focused on volume optimization and it is an individual task.

Instructions

- Each student will be provided with rectangular sheets of paper with the objective of maximizing volume.
- Before commencing the practical task, students are expected to conduct research on relevant theories and grasp the underlying principles.
- After finding the proper solution, then perform the rectangular sheet activity.

Following the completion of the activity, students will present their findings to the class, fostering discussion and further understanding of volume optimization concepts.

Week 3 Discussion ([Rubric](#))

In this discussion, let's explore how differentiation, a fundamental concept in calculus, is applied to solve real-world problems. Identify a practical example where differentiation is used—such as optimizing a business process, analyzing rates of change in physics, or modeling population growth. Briefly explain how differentiation is applied in your example, the results it helps achieve, and any limitations or challenges you encountered.



Week 4

Assignments to complete this week:

- Reading: Calculus: Early Transcendentals (p. 371-434)
 - Introduction to integral calculus
 - Riemann method to find sum
 - Definite and Indefinite Integrals
 - Rules of integration
 - Integration by substitution and integration by parts
- Video:
 - [Approximating Area Under a Graph Using Rectangles](#)
 - [The Antiderivative](#)
 - [The Definite Integral](#)
 - [Basic Antiderivatives of Trigonometric Functions](#)
 - [Integration by Substitution](#)
 - [Integration by Parts: The Basics](#)

Week 4 Live Class Activities ([Rubric](#))

This group activity will be conducted during the live class session and requires individual participation for integral problems solving. Here are the instructions:

Instructions

- The instructor will provide few integration problems to be solved and any other guidelines such as tools, time frame etc.
- Each student needs to solve the problem as guided by the instructor.

Following the completion of the activity, students will have the opportunity to present their findings to the live class.

Week 4 Assignment ([Rubric](#))

For this assignment, students are required to solve questions by using pen and paper from the textbook. Complete the odd numbered questions (1-40) from review exercise of chapter 5, pages 429-430.

Submission Guidelines: Scan your completed solutions, compile them, and upload them to the portal in PDF format.



Week 5

Assignments to complete this week:

- Reading:
 - Area under and between the Curve
 - Volume of revolution
 - Average Value of Function
 - Applications of integral calculus in economics and science
 - Finding probability
- Videos:
 - [Determining Area Between Two Curves - Integration Application](#)
 - [Volume of Revolution - The Disk Method](#)
 - [Ex 1: Average Value of a Function](#)
 - [Ex 1: Application of Definite Integration](#)
 - [Ex: Integration Application - Present Value for Business](#)
 - [Integration Application: Displacement and Distance](#)
 - [Probability Density Functions](#)

Week 5 Live Class Activities ([Rubric](#))

The objective of this live class activity is to understand the real-world integral applications, enhancing problem-solving skills, and fostering teamwork and critical thinking.

Instructions

- The instructor will divide the class into different groups based on the class size.
- Each group needs to perform the following activities:
 - estimate the area of a lake by plotting its map on graph paper and dividing it into geometric slices.
 - Calculate the area using integrals.

Following the completion of the activity, each group will present their findings to the live class.

Week 5 Assignment ([Rubric](#))

For this task, students are required to search online to locate suitable examples demonstrating the practical application of integration. Once you've found the proper example, apply an integral process to find the results. Then, compile your findings into a concise 2-3 page report following APA format guidelines. Your report should include sections such as title, problem statement,



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tools (if applicable), methodology and calculations, findings and results, limitation of the solution, and references.



Week 6

Assignments to complete this week:

- Reading:
 - Understanding of Multivariable functions
 - Limits and Continuity of multivariable functions
 - Partial Derivatives and its applications
 - Chain Rule for Multivariable Functions
 - Maximum and Minimum Values
 - Double integral and its applications.
- Video:
 - [Introduction to Functions of Two Variables](#)
 - [Limits of Functions of Two Variables](#)
 - [First Order Partial Derivatives](#)
 - [The Chain Rule for Functions of Two Variable with One Independent Variable](#)
 - [Absolute Extrema of Functions of Two Variables](#)
 - [Introduction to Double Integrals and Volume](#)

Week 6 Discussion ([Rubric](#))

Discuss at least four applications of multivariate calculus within the information technology sector. Select a specific use case and delve into how multivariate calculus is employed to address challenges within this context. Provide solutions derived from the application of multivariate calculus in your discussion.

Week 6 Assignment ([Rubric](#))

Complete following tasks using pen and paper with narrative explanation.

1. You're the engineer in charge of designing a rectangular solar panel to maximize its energy output. The panel's length and width are denoted by x and y respectively, and its energy output (in megawatts) is modeled by the function:

$$E(x,y) = 2.5xy - 0.04x^3 - 0.02y^3$$

Constraints:

- The available space allows for a maximum length of 5 meters and a maximum width of 3 meters.
- The cost of production increases as the area of the panel grows, represented by a cost function:

$$C(x, y) = 100x + 5y^2 + 3y^2$$

- a. Find the partial derivatives of $E(x, y)$ with respect to x and y .
- b. Use the concept of critical points to determine the dimensions (length and width) of the panel that maximize energy output within the given constraints.



- c. Calculate the maximum attainable energy output and the corresponding production cost using the optimal dimensions.
2. Consider a region R in the xy -plane bounded by the curves $y=x^2$ and $y=4$. You are required to find the volume of the solid that lies above the region R and below the surface $z=6-x^2$.
- Sketch the region R in the xy -plane.
 - Set up the double integral for computing the volume of the solid.
 - Evaluate the double integral to find the volume.

Submission guidelines: Scan your completed solutions, compile them, and upload them to the portal in PDF format. Make sure that the solution format should align with the topics covered and discussion in lectures.



Week 7

Assignments to complete this week:

- Reading:
 - Introduction to Calculus using Python
 - Introduction to sympy and numpy libraries in python
 - Limit using Python
 - Numeric derivative using Python
 - Symbolic Derivative using Python
 - Numeric integral using Python
 - Symbolic integral using Python
- Video:
 - [1st Year Calculus, But in PYTHON](#)
 - [Derivatives In PYTHON \(Symbolic AND Numeric\)](#)
 - [Integration in PYTHON \(Symbolic AND Numeric\)](#)

Week 7 Live Class Activities ([Rubric](#))

Organize the class into distinct groups according to its size. Each group will be tasked to set up an environment, use python libraries for calculus. Engage in the following activities:

- Set up the environment
- Explore different python libraries for calculus such as NumPy, SymPy
- See the output and analyze

Students need to present their findings to the live class.

Week 7 Assignment ([Rubric](#))

Solve the following problems using python and submit it with a narrative explanation of process and result.

1. You are a manager at a manufacturing plant, and you want to optimize the production of a certain product. The cost function for producing x units of the product per week is given by $C(x) = 10000 + 5x + 0.001x^2$, and the revenue function is $R(x) = 50x$

- a. Determine the profit function $P(x)$.
- b. Find the production level that maximizes profit.
- c. What is the maximum profit?



2. The population of bacteria after t hours is modeled by:

$$P(t) = \frac{30000}{1 + 500e^{-0.08t}}$$

where $0 < t \leq 200$.

Find the following:

- a. $P'(t)$
 - b. The rate of increase of the bacteria after 30 hours
 - c. The time at which the population is increasing by 200 bacteria per hour.
 - d) The time at which the bacteria population is growing the fastest
3. The time spent waiting in line at a certain bank is modeled by an exponential density function with a mean of 10 minutes.
- a. What is the probability that a customer is served in the first 4 minutes?
 - b. What is the probability that a customer has to wait more than 12 minutes?
 - c. What is the probability that a customer has to wait between 8 to 12 minutes?
 - d. What is the median waiting time?

Week 8

Assignments to complete this week:

- Explore the various applications of calculus.
- Application of calculus using python

Week 8 Live Class Activities ([Rubric](#))

In this live group activity, students will delve into the practical applications of calculus through Python programming. Following a concise review of calculus principles, the session will entail the following steps:

- Students will identify real-world scenarios where calculus finds application.
- Through coding demonstrations and collaborative efforts, students will learn to translate these scenarios into mathematical functions, employing Python libraries to execute calculus operations.
- The instructor will facilitate group formation based on class size and assist students in selecting appropriate tools and real-world scenarios, if necessary.

Each group will subsequently present their problem-solving approach and outcomes to the class.

Week 8 Exam

Limit, Differential, and Integral Calculus Exam: This 75-minute exam will evaluate the material covered in weeks 1 through 7 and will take place during a live class under the supervision of a proctor.